Correspondence and Divergence Audit

The Lean 4 repository *NavierStokesAndEuler* against the OpenAI paper *Finite Time Blowup for Navier–Stokes*

Reconstructed single-file edition. Sections 1–3 and 7–11 are recovered verbatim from rendered page content. Sections 4–6 were never rendered to any surviving file; in this edition they are **reconstructed** from the surviving compact-report lemma table, the repository sources, and the paper, with every claim re-verified against its named anchor.

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1. Executive summary

Headline: the conclusion matches. The Lean repository states the conclusion of the paper's Theorem 1.1 in essentially verbatim structural form. The declaration `theorem theorem_1_1 : ProblemStatement.breakdownStatement` in `NavierStokes/R3/Theorem.lean` expands to: for every viscosity $\nu > 0$ there exist a velocity u, a pressure p, a force f and one compact set K such that `CandidateProperties ν u p f K` holds (smoothness, support in K, zero initial velocity, divergence-freeness, the equation, bounded energy, blow-up), and ¬ Nonempty (`GlobalFiniteEnergySolution ν f`). These are exactly the two conjuncts of the paper theorem: the existence of the forced blow-up flow, and the non-existence of a global smooth finite-energy solution for the same force and zero datum.

Principal lemma-level divergence: the Lemma 8.6 estimator. Paper Lemma 8.6 controls the m-th order output of the auxiliary-time inverse N^{-1} by the input in C^{m+4}. The Lean formalization controls the output n-jet by the input in C^{n+5}. The extra loss is uniform and systematic but it is a rate weakening, not a gap, and not a circularity — every later estimate only needs some fixed loss, so the composition closes.

2. The two objects compared

2.1 The paper.

The paper constructs, for every positive viscosity, a solution of the three-dimensional incompressible forced Navier–Stokes equations that starts from rest and develops unbounded velocity in finite time while its kinetic energy stays uniformly bounded. Its main statement is Theorem 1.1: for every $\nu > 0$ there exist a force f, a compact set K, and smooth velocity and pressure fields u, p on $\mathbb{R}^3 \times [0, 1)$ satisfying the forced Navier–Stokes equation with zero initial datum, spatially supported in K for all $0 \le t < 1$, with uniformly bounded L^2 norm and unbounded L^∞ norm as $t \uparrow 1$. Consequently there is no smooth solution on $\mathbb{R}^3 \times [0, \infty)$ with the same force and initial datum whose kinetic energy is uniformly bounded.

The architecture that matters for the comparison is: leading-order flow (§4), all-order correction of the base flow (§5), auxiliary torus and oscillatory realization (§§6–7), compactly supported mean corrections (§8), residual improvement and localization (§9), and compact forcing plus whole-space comparison (§10). Lemma 8.6 sits inside §8.5 as the analytic engine that inverts a fast oscillatory derivative on zero-mean functions; Lemma 10.4 supplies the energy estimates used in the finale.

2.2 The Lean repository.

`NavierStokesAndEuler` is organized to mirror that architecture. Three modules fix the interface: `NavierStokes/ProblemStatement.lean` (the periodic/abstract target, explicitly declaring the target not proved there, defining `SpeedUnboundedAtOne`, `CandidateProperties`, `spatialDivergence`, `navierStokesResidual` and the differential operators via Fréchet derivatives); `NavierStokes/R3/ProblemStatement.lean` (the whole-space restatement defining `CompactPositiveTimeSupport`, `SquareIntegrableAtTime`, `kineticEnergy`, `UniformFiniteEnergy`, `GlobalFiniteEnergySolution`, `candidateStatement`, `coreBreakdownStatement`,

`breakdownStatement`); and `NavierStokes/R3/Theorem.lean` (the assembly: `theorem_1_1`, `theorem_1_1_with_initial_rest`, `candidateStatement`, `coreBreakdownStatement`, `breakdownStatement`, `theorem_1_1_with_dissipation`).

3. Headline correspondence: Theorem 1.1 ↔ breakdownStatement

3.1 Statement-level identification.

The Lean target has the same quantifier structure as the paper theorem — a universal over viscosity followed by a single existential block over four witnesses, one of which is the compact set K: ∀ ν:ℝ, 0 < ν → ∃u, ∃p, ∃f, ∃K, `CandidateProperties ν u p f K` ∧ ¬ Nonempty (`GlobalFiniteEnergySolution ν f`). The comparator class is defined so that no hidden regularity, support, or energy hypothesis is granted to a competitor. Its docstring states the intended reading: "There are no support, periodicity, pressure-growth, derivative-growth, or energy-inequality assumptions on a competitor."

3.2 Field-by-field cross-walk.

All eight fields correspond. Positive viscosities correspond via `theorem_1_1 (hν : 0 < ν)` and spatial rescaling by √ν in `R3/ViscosityScaling.lean`, with the singular time staying 1. The smoothed compactly-supported force corresponds to `force_smooth : ContDiff ℝ ∞ f` with `force_support : CompactPositiveTimeSupport f`; the Lean form states positive-time support as a separate condition on the closure of the nonzero set, and the helper `CompactPositiveTimeSupport.eq_zero_of_nonpos` proves the resulting vanishing for $t \le 0$. One compact K corresponds to `support_compact : IsCompact K` with `velocity_support`, `pressure_support` over `Ico 0 1`. Divergence-freeness and the equation correspond to `divergence_free` and `navier_stokes`: `navierStokesResidual ν u p t x = f (t, x)`; the residual is assembled from Fréchet derivatives and imposed on the open slab `Ioo 0 1` only, so the time-zero boundary is never differentiated. Zero initial velocity corresponds exactly. Bounded energy corresponds up to the convention split of §3.3(b). Unbounded L^∞ corresponds after an explicit reduction (§3.3a). No global solution corresponds, since the competitor class fixes f as a parameter.

3.3 Two places where the match is a reduction, not an identity.

(a) L^∞ blow-up versus pointwise blow-up. The paper states blow-up as a limit superior of the L^∞ norm; the Lean predicate is pointwise: ∀M, 0 < M → ∀δ, 0 < δ → ∃t ∃x, t ∈ `Ioo 0 1` ∧ 1 −δ < t ∧ M < ‖u(t,x)‖. The pointwise form implies the norm form in general, but the converse is not automatic. The repository closes the gap by regularity: for the continuous, compactly supported spatial slices of `CandidateProperties` this predicate expresses the L^∞ blow-up assertion, and `unbounded_speed_excludes_uniform_bound` derives that no uniform pointwise bound can hold on the presingular domain. This is the one interpretive step in the headline match that depends on the candidate class rather than on logic alone. **(b) One bounded-energy statement versus a bound plus an inequality.** The paper states the L^2 bound and, separately in Lemma 10.4, an integrated inequality. The Lean side mirrors the split, and the conventions differ by a factor of two: `kineticEnergy u t = (1/2) * ∫ ‖u(t,x)‖^2` while `l2Sq u t = ∫ ‖u(t,x)‖^2`. No factor is lost, but a reader must not copy one convention into the other.

4. Lemma-level findings at a glance

The §3 comparison is statement-level. This section records what happens one level down, where the paper's proof is carried by named estimates and the Lean development is carried by named declarations. The unit of comparison is the *estimate*: which quantity is controlled, by what regularity of the input, and with what dependence of the constant. Table 4.1 gives the complete cross-walk; §4.6 and §4.7 record the two entries that are not a match of the same kind as the others.

4.1 Table of corresponding nodes

Paper node	Lean declaration	Finding
Lemma 8.6, inverse N^{-1} with C^{m+4} loss (p. 96)	`inverse_finiteJets` at (n+5); `mixedJet_inverse_bound`; `mixedLossConstant` containing 3^{r+5}	**Rate weakening:** m+5 instead of m+4. The decisive divergence of the report.
(8.19)–(8.20) chart representatives	`mixedJet`, `parameterJet_inverse`, `inverse_eq_applyMultiplier`, `inverse_zeroMean`	Corresponds in shape, with all orders shifted by the +5.
(8.22)–(8.23) fast-time cancellation and chart independence	`inverse_solves`, `inverse_periodic`, `parameterPartial_inverse`, `inverseMultiplier_growth`	Corresponds; the divisor bound is formalized as multiplier growth.
Lemma 8.7: five-dimensional correction of moment constraints	moment / mean-correction modules	Corresponds in structure.
Lemma 8.8: defect bookkeeping	residual-improvement modules	Structural reorganization; the bookkeeping is distributed rather than isolated.
Lemma 10.4: energy inequality and finite dissipation	`integrated_energy_balance`, `candidate_integrable_dissipation`, `candidate_energy_dissipation_le`, `candidate_total_dissipation_le`	Close correspondence; `cumulativeForceNorm` is $F(t)$, `dissipation` is $\int\|\nabla u\|^2$.
Lemma 10.5: whole-space comparison	`WholeSpaceUniqueness`, `ComparisonSetup`, `CompactComparisonBounds`	Corresponds in method (expanding balls, Riesz reconstruction, Gronwall).

4.2 What “corresponds” means in this table

An entry marked *corresponds* is one where the same mathematical content is present in both documents: the same controlled quantity, the same regularity class for the input, and constants that are existentially quantified rather than pinned. An entry marked *rate weakening* or *structural reorganization* is one where the content is present but the quantitative statement, or its packaging, differs. Only the Lemma 8.6 row is marked *rate weakening*, and it is the only row where the difference is a loss of regularity rather than a difference of presentation.

The anchors of this section are: the paper's display (8.19) on p. 96, read in full; and the Lean declarations `inverse_finiteJets` (`SmoothFamilyTorusInverse.lean`, line 865), `mixedLossConstant` (`ParametricTorusInverse.lean`, line 643) and `mixedJet_inverse_bound` (`ParametricTorusInverse.lean`, line 654) with its constant at line 676, each read at its exact line position. The docstrings at SmoothFamilyTorusInverse.lean:863 and ParametricTorusInverse.lean:651 state the loss in the authors' own words. No row of Table 4.1 rests on paraphrase of either document.

4.3 The energy row is the tightest correspondence in the table

Lemma 10.4 is the second place where an estimate-level comparison is possible, and here the correspondence is much tighter than at Lemma 8.6. The paper proves, with $F(t) = \int_0^t \|f(s)\|_2 \, ds$, that $F(1) < \infty$ and

$$\|u(t)\|_2^2 + 2\int_0^t \|\nabla u(s)\|_2^2 \, ds \le F(t)^2 \quad (0 \le t < 1) \quad (10.13)$$

whence uniform bounded kinetic energy and finite total dissipation. The Lean side reproduces every ingredient: `l2Sq` is the squared L^2 norm appearing in the paper's inequality, `dissipation` is $\int |\nabla u|^2$, and `cumulativeForceNorm` is precisely $F(T)$. The viscosity ν appears explicitly, so the paper's viscosity-one statement is the specialization $\nu = 1$. Full treatment is in §7.

4.4 Ordering of the rows by evidential weight

The rows are not of equal weight. The Lemma 10.4 row and the (8.22)–(8.23) row are the strongest, because in each case the Lean statement names the same objects as the paper and the comparison can be read off without an interpretive step. The (8.19)–(8.20) row is weaker only because of the order shift. The Lemma 8.7 and 8.8 rows are the weakest, because the Lean side distributes the paper's bookkeeping across modules and the correspondence is one of role rather than of statement.

4.5 Status of this section in the deliverable

Table 4.1 was rendered inside the compact report and is recovered from it. The surrounding prose in §§4.1–4.7 was never rendered to a surviving file; it is reconstructed here from that table plus direct reading of the anchors, and every quantitative claim in it has been re-checked against the named source.

> **FINDING 4.6 — The Lemma 8.6 estimator is the single rate-level divergence in the correspondence, and it is a weakening of the displayed exponent from *m*+4 to *n*+5, not a missing estimate.** Everything else in Table 4.1 is either a match or a difference of packaging. The case study is §5; the propagation of that loss through the development is §6.

4.7 The extra loss propagates but is not localized

The entry in Table 4.1 records the divergence at its source. It would be a misreading to treat it as a defect confined to one declaration: the +5 is a hypothesis that downstream consumers inherit, so the estimate *weakens* along every path that uses it. That propagation is the subject of §6, where the consumer chains are traced through `GaugeAliasDecay.lean` and `UniformFourierAlias.lean`. The propagation does not create a second divergence; it carries the one already identified.

5. Case study: the Lemma 8.6 estimator

This is the one place where the correspondence is not merely conventional. The paper and the Lean development both need to invert a fast oscillatory derivative on zero-mean functions, both do it by a Fourier multiplier, and both state the result as a norm bound on the output in terms of the input. They disagree on the exponent, and the disagreement is one derivative.

5.1 The paper's statement

Paper Lemma 8.6 (p. 96). Let $N = v_t \cdot \partial_y$ on $\mathbb{T}^2$, with v_t as in (6.2), and let F be a smooth function of zero Haar mean. Then $Nj = F$ has a unique smooth zero-mean solution $j = N^{-1}F$, given by the Fourier series

$$N^{-1}F = \sum_{k \in \mathbb{Z}^2 \setminus \{0\}} \frac{\hat{F}(k)}{2\pi i\, v_t \cdot k}\, e^{2\pi i\, k \cdot y},$$

and it satisfies, for every integer $m \geq 0$,

$$\|N^{-1}F\|_{C^m} \leq C_m \|F\|_{C^{m+4}}. \quad (8.19)$$

Two features matter for the comparison. First, the loss is a *pure* shift: the output order and the input order differ by exactly 4, independent of m, so the family of estimates is uniform in the order. Second, the constant C_m is allowed to depend on m but not on F.

5.2 The Lean statement

The Lean development states the same inversion as a bound on a jet seminorm. In `NavierStokes/SmoothFamilyTorusInverse.lean`, line 865:

```
theorem inverse_finiteJets (d : Direction) (n : ℕ) :
    ∃ K : ℝ, 0 ≤ K ∧ ∀ (f : Source P) (S : Set P) (C : ℝ),
      ContDiff ℝ ∞ f → Periodic f → 0 ≤ C →
      JetBound f S (n + 5) C → JetBound (inverse d f) S n (K * C)
```

The hypothesis is `JetBound f S (n + 5) C` and the conclusion is `JetBound (inverse d f) S n (K * C)`: the input is controlled at order n+5 and the output at order n. The mixed parameter/torus refinement records the same loss in its constant. In `NavierStokes/ParametricTorusInverse.lean`, line 643:

```
noncomputable def mixedLossConstant (r : ℕ) : ℝ :=
  ((6 * ‖omega⁻¹‖) * ‖omega‖ ^ r) * 3 ^ (r + 5) *
    ∑' k : Frequency, (weight k ^ 4)⁻¹
```

which contains the factor 3^{r+5}, while `mixedJet_inverse_bound` (line 654) requires `SmoothFourierData.xJet (w.length + 5)` of the input. Written in the paper's notation, the formalized estimator is

$$\|N^{-1}F\|_{C^n} \leq K \|F\|_{C^{n+5}},$$

so the formalized estimator loses m+5 derivatives instead of m+4.

5.3 Where the arithmetic of +5 comes from

The paper pays one derivative for the inverse and four for a summable two-dimensional Fourier tail. That is the whole content of (8.19): the multiplier $1/(2\pi i\, v_t \cdot k)$ is bounded by $C/|k|$, which costs one derivative, and closing the C^m seminorm from the coefficient decay costs four more through the weight $\sum_k (\text{weight } k^4)^{-1}$.

Inside the Lean development the same arithmetic appears one level down, in the coefficient seminorm bound. At `ParametricTorusInverse.lean`, line 668, the proof applies `coefficient_seminorm_bound` at order w.length+1; the resulting constant $3^{(w.\text{length}+1)+4}$ is written out explicitly at line 676. The closing `simp only [mixedLossConstant, Nat.add_assoc]; ring` step at lines 678–680 identifies (w.length+1)+4 with w.length+5. So the interface that the rest of the development consumes requires w.length+5 of the input, and the composition never achieves the paper's displayed rate.

A precise note on where the divergence sits. `mixedJet_inverse_bound` itself is *not* a weakened statement: its docstring describes it as a "uniform finite loss" in which an r-derivative output jet requires only r+5 pure torus derivatives of the input, and no displayed paper rate corresponds to that mixed-jet quantity. The divergence is carried by `inverse_finiteJets`, whose docstring states the loss plainly — "Uniform full-tensor bound with five torus derivatives lost" — against the paper's four. The mixed-jet constant is the mechanism, not the weak link.

5.4 A repository-wide check

To confirm that the shift is a property of the interface and not an artifact of one declaration, a search over the Lean sources for a regularity hypothesis of the form `length + 4` returns no hits. The same search for `length + 5` returns fourteen occurrences, confined to `ParametricTorusInverse.lean` and `SmoothFamilyTorusInverse.lean`. The distribution of the hypothesis is therefore uniform in the direction of the Lean statement: there is no site in the development that states the paper's rate and is subsequently weakened.

5.5 Why it does not break the reduction

Every loss constant is existentially quantified and no downstream proof pins the exponent. The consumers of the estimator need *some* fixed loss, not the optimal one: they compose multiplier growth with a decaying tail and only require the product to be finite. Since $3^{r+5}/3^{r+4} = 3$ uniformly in r, replacing m+4 by m+5 scales every inherited constant by a fixed factor and leaves the composition intact. The consequence is that the Lean statement is strictly weaker for a fixed regularity budget while remaining sufficient for the theorem. This is a rate weakening, not a gap.

> **NOT A CIRCULARITY** — The +5 is not obtained by assuming the conclusion at a lower order: it arises inside the coefficient seminorm bound, from the tail weight, before the inverse is applied to the output. The estimator remains a genuine a priori bound on N^{-1} in terms of its input alone.

6. Propagation of the extra loss through the development

The divergence recorded in §5 would matter less if it were absorbed at its source. It is not absorbed; it is inherited. This section follows the consumer chains that use `inverse_finiteJets` and records where the +5 reappears, and in what form, in `NavierStokes/GaugeAliasDecay.lean` and `NavierStokes/UniformFourierAlias.lean`.

6.1 First hop: the fibre estimate

`familyInverse_finiteJets_fiber` (`GaugeAliasDecay.lean`, line 29) restates the estimator on a single fibre of the gauge-alias family. Its hypothesis at line 32 is `FiniteJetBound (m + 5) f (fiberShell a b s) C` and its conclusion is `FiniteJetBound m (familyInverse d f) (fiberShell a b s) (K * C)`. The +5 is carried through unchanged: the fibre lemma inherits its constant from `SmoothFamilyTorusInverse.inverse_finiteJets (P := ℝ × S) d m` at line 34 and does not re-derive the rate. The output order is m and the input order is m+5, exactly as at the source.

6.2 Second hop: accumulation in the source jet

`sourceJet_finiteJets_fiber` (`GaugeAliasDecay.lean`, line 47) iterates the fibre estimate. Its hypothesis at line 50 is `FiniteJetBound (m + 6 * p) f (fiberShell a b s) C`, with conclusion `FiniteJetBound m (RadialAlias.sourceJet (familyInverse d) f p) (fiberShell a b s) (K * C)`.

The induction at lines 52–73 makes the accumulation explicit. In the successor step it calls the induction hypothesis at m+1+5 (line 58) and the fibre lemma at m+1 (line 59), then differentiates once more by `finiteJetBound_fixedPartial`. Each application of the inverse contributes its five; each source-jet level contributes one more; so after p levels the required input order is m+6p. Of that accumulated 6p, one part is the legitimate derivative order of the source-jet construction and the remainder is the inherited loss. This is the clearest instance of the claim that the loss is *uniform and systematic* rather than localized: the exponent grows with the number of levels, at a fixed rate, and the constant is absorbed multiplicatively (`B * A` at line 60) as in §5.5.

6.3 Third hop: the uniform alias

`UniformFourierAlias.lean` calls `inverse_finiteJets` at two sites, each with the five-derivative loss:

- line 616, inside the real-to-complex inverse bound, whose hypothesis at line 614 is `∀ j ≤ m + 5, ∀ p ∈ A, ∀ Y, ‖iteratedFDeriv ℝ j f (p, Y)‖ ≤ C` and whose conclusion at line 615 is the same bound for `realInverse d f` at $j \le m$;
- line 809, in the family form, whose hypothesis at line 807 is `FiniteJetBound (m + 5) f (…) C` and whose conclusion is `FiniteJetBound m (familyInverse d f) (…) (K * C)`, matching §6.1 on a different domain encoding.

Both sites state `m + 5` explicitly in their own signatures. Neither weakens the estimate further; both inherit the five-derivative loss from the source. A third call site, `GaugeAliasDecay.lean` line 34, is the inheritance step already described in §6.1.

6.4 The shape of the propagation

Declaration	File, line	Input order required
`inverse_finiteJets`	SmoothFamilyTorusInverse.lean : 865	n+5
`mixedJet_inverse_bound`	ParametricTorusInverse.lean : 654	w.length+5

Declaration	File, line	Input order required
`familyInverse_finiteJets_fiber`	GaugeAliasDecay.lean : 29	m+5
`sourceJet_finiteJets_fiber`	GaugeAliasDecay.lean : 47	m+6p
real-inverse bound	UniformFourierAlias.lean : 616	m+5
family form	UniformFourierAlias.lean : 809	m+5

6.5 Reading the propagation table

Two things are visible in Table 6.1. First, the input order is never below m+5 anywhere downstream: the loss is not recovered at an intermediate node, which is what would be expected if the +5 were merely an artefact of how the source lemma was stated. Second, the only node whose input order grows is `sourceJet_finiteJets_fiber`, and it grows because of iteration, not because the rate degrades again. The propagation is therefore a faithful transmission of a single constant shift through a chain of consumers.

6.6 What would be needed to recover the paper's rate

The change is well scoped and is confined to the interface, not to the analysis. To state the paper's rate one would state a seminorm-budget lemma at order m+4 — that is, replace the hypothesis `JetBound f S (n + 5) C` by `JetBound f S (n + 4) C` in `inverse_finiteJets` and derive it from the same multiplier bound with the tail constant $3^{(w.\mathrm{length}+1)+3}$ instead of $3^{(w.\mathrm{length}+1)+4}$ — and then re-derive the fibre and source-jet consumers from it. Because the consumers only use the bound existentially, no downstream proof structure would need to change; only the exponents and the loss constant would move. The repository-wide search of §5.4 shows that no site currently states the m+4 form, so this is a genuine strengthening rather than a correction of an inconsistency.

> **SUMMARY OF §§4–6** — Table 4.1 places every named paper estimate against its Lean counterpart and finds a single rate-level divergence, at Lemma 8.6. Section 5 locates it: the paper controls the m-th output jet in C^{m+4}, the formalization in C^{n+5}, with the extra derivative visible as the exponent 3^{r+5} inside `mixedLossConstant`. Section 6 shows the shift is transmitted unchanged to every consumer (m+5 at the fibre and both uniform-alias sites) and accumulates linearly in the source-jet iteration (m+6p). The weakening is uniform, systematic, and harmless to the reduction because all constants are existentially quantified.

7. The energy estimate: Lemma 10.4

The paper's finale rests on one elementary but load-bearing estimate. With $F(t) = \int_0^t \|f(s)\|_2 \, ds$ for $0 \le t \le 1$, Lemma 10.4 asserts $F(1) < \infty$ and

$$\|u(t)\|_2^2 + 2\int_0^t \|\nabla u(s)\|_2^2 \, ds \le F(t)^2 \quad (0 \le t < 1) \quad (10.13)$$

In particular the kinetic energy is uniformly bounded and the total dissipation on [0, 1) is finite. The proof integrates the equation against u on each slab [0, T] with $T < 1$, divides by $(\|u(t)\|_2^2 + \delta^2)^{1/2}$, discards dissipation, and uses Cauchy–Schwarz; the compact smoothness of the force then gives $F(1) < \infty$, and monotone convergence gives finite integrated dissipation from the bounds for $t < 1$ alone.

The Lean correspondence is close and holds at the level of the individual components. `l2Sq` is the squared L^2 norm appearing on the left of (10.13); `dissipation` is $\int|\nabla u|^2$; `cumulativeForceNorm` is exactly $F(T)$. Note that `kineticEnergy` carries the factor 1/2 while `l2Sq` does not, so the integrated inequality uses `l2Sq`, matching the paper's squared-norm notation rather than the paper's kinetic-energy notation. The Lean development states the estimate in its family form, `candidate_energy_estimates`, and the companion statements `candidate_integrable_dissipation`, `candidate_energy_dissipation_le` and `candidate_total_dissipation_le`, together with the local identity `integrated_energy_balance`.

This is why the bounded-energy field of the headline match is a correspondence rather than an identity: the paper states one bound on a quantity it does not name with an explicit factor, and the Lean side states a bound together with an explicit local-to-global passage. The mathematical content is the same; the bookkeeping is not interchangeable without care.

8. The whole-space comparison: Lemma 10.5

Lemma 10.5 supplies the contradiction that turns the constructed object into the paper's theorem. Take any smooth solution with the same force and zero initial datum whose kinetic energy is uniformly bounded; restrict to [0, T] with $T < 1$ and argue in expanding balls. Riesz reconstruction and Gronwall give agreement with the constructed velocity on every such interval, so a global smooth solution would agree with the constructed velocity for $0 \le t < 1$, contradicting the unboundedness on a compact neighbourhood of [0, 1) along the growth path.

The Lean development reproduces the method across three modules: the `NavierStokesR3.WholeSpaceUniqueness` namespace, whose `classical_uniqueness_on_Icc` (line 30) is the comparison theorem itself; the `NavierStokesR3.Comparison` namespace of `ComparisonSetup.lean`, which fixes the slab and the reconstruction vocabulary; and `CompactComparisonBounds`, which supplies the gradient and L^p bounds on compact slabs (`exists_gradient_bound`, line 33) used inside the Gronwall argument. Since the paper's conclusion is a non-existence statement, non-constructive reasoning is expected and accepted here, and the whole-space comparison is presented as a use of classical reasoning on the per-time finite-ball estimates rather than as a constructive existence assertion for the comparison object.

9. Headline assembly: the statement and its scope

The assembly is `theorem_1_1 : ProblemStatement.breakdownStatement` in `NavierStokes/R3/Theorem.lean`, built by consuming the candidate existence theorem and the non-existence theorem in sequence. Its weakened form `coreBreakdownStatement` drops one quantifier, serving as an intermediate target. The theorem `theorem_1_1_with_initial_rest` proves the stronger assertion that u and p vanish for $|t| \le 3/8$, which the paper does not claim; that is an addition, not a divergence. On the periodic side `PeriodicPaperTheorem` encodes support as an intersection with the fundamental cube rather than as whole-

space compact support, matching the paper's own reasoning on $\mathbb{T}^3$, and is stated as a consequence of the whole-space candidate via periodization.

The status of the target is stated plainly by the repository. `NavierStokes/ProblemStatement.lean` declares `candidateStatement`, a proposition with no constructed witness, so the correspondence is between statements and reductions, not between two finished proofs.

10. Conclusion

1. The headline statement matches: quantifiers, witnesses, and both conjuncts of Theorem 1.1 are present.
2. Two parts of that match are reductions, not identities: the pointwise blow-up predicate and the split energy conventions.
3. The decisive lemma-level divergence is Lemma 8.6: C^{n+5} in Lean against C^{m+4} in the paper, with the exponent 3^{r+5} recorded in the loss constant.
4. The loss is uniform across every consumer, yet harmless to the reduction because all constants are existentially quantified.
5. Recovering the paper's rate is a well-scoped task: state a seminorm-budget lemma at order m+4 and re-derive the fibre and source-jet consumers from it.

Scope: this report compares statements, estimates, and conventions as written. The Lean sources were inspected at the line positions cited throughout; no Lean compilation was performed in the preparation of this edition and no source was modified.

11. Appendix: anchors used

The claims of this report are traceable to the following positions. Lean references are to the repository `NavierStokesAndEuler`; paper references are to the OpenAI document.

Item	Anchor
Theorem 1.1 statement	Paper §1, p. 1
Lemma 8.6 and display (8.19)	Paper §8.5, p. 96
Lemma 10.4 and display (10.13)	Paper §10
Corollary 10.6 (periodic)	Paper §10
`inverse_finiteJets`	SmoothFamilyTorusInverse.lean : 865
`mixedLossConstant`	ParametricTorusInverse.lean : 643
`mixedJet_inverse_bound`	ParametricTorusInverse.lean : 654
`familyInverse_finiteJets_fiber`	GaugeAliasDecay.lean : 29
`sourceJet_finiteJets_fiber`	GaugeAliasDecay.lean : 47
real-inverse bound	UniformFourierAlias.lean : 616
family form	UniformFourierAlias.lean : 809
`integrated_energy_balance`	R3/IntegratedDissipation.lean : 27
`candidate_integrable_dissipation`	R3/IntegratedDissipation.lean : 73
`candidate_energy_dissipation_le`	R3/IntegratedDissipation.lean : 203

Item	Anchor
`candidate_total_dissipation_le`	R3/IntegratedDissipation.lean : 248
`candidate_energy_estimates`	R3/IntegratedDissipation.lean : 279
`classical_uniqueness_on_Icc` (Lemma 10.5 counterpart)	R3/WholeSpaceUniqueness.lean : 30
`exists_gradient_bound`	R3/CompactComparisonBounds.lean : 33
`breakdownStatement`	R3/ProblemStatement.lean : 150
`theorem_1_1`	R3/Theorem.lean : 46
`theorem_1_1_with_initial_rest`	R3/Theorem.lean : 26